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Water Needs, Water Insecurity, and Human Biology

Asher Y. Rosinger

Department of Biobehavioral Health and Department of Anthropology, Pennsylvania State University, University Park, Pennsylvania, USA; email: arosinger@psu.edu

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Keywords

water insecurity, human biology, hydration, thirst, coping strategies, embodiment, climate change

Abstract

Water links the environment, culture, and biology. An integrative approach is needed to attain a complete picture of how water affects human biology due to its inherent interdisciplinary nature. First, this review describes advances in human water needs, thirst, and hydration strategies from a bio-cultural perspective. Second, it provides a critical appraisal of the literatures on water insecurity (WI) experiences and coping strategies used to mitigate WI to illustrate how they intersect to affect human biology through the embodiment framework. Deviations from water needs and heightened WI can alter hydration and coping strategies, which have implications for a suite of psychological and physiological outcomes. These disruptions are embodied in cellular damage, dehydration, nutrition, stress, mental health, cognitive impairment, aging-related effects, cardiometabolic health, and kidney function. Disrupting forces such as lifestyle changes and climate change have important implications for water needs, WI, coping and hydration strategies, and the embodiment of each.

INTRODUCTION

Water is the most basic human need, yet it is among the most complex biocultural phenomena as it links the environment, culture, and biology (Rosinger 2020). Water needs, use, and security shape and are shaped by all facets of life—cultural, dietary, political, and environmental. There has been burgeoning research interest on water-related issues in anthropology and other fields due to water crises, climate change, and the declaration of water as a basic human right and its inclusion in the United Nations' sustainable development goals. Two prior *Annual Review of Anthropology* articles used political, economic, and environmental anthropology approaches to discuss water (Ballesterio 2019, Orlove & Caton 2010). Yet, globally, two billion people still lack access to safely managed drinking water (WHO & UNICEF 2021), which has tremendous health and human biology implications. An integrative approach linking physiological water needs to water insecurity (WI) is needed to attain a complete picture of how water affects human biology due to its inherent interdisciplinary nature.

There has been little integration of the water needs and WI literatures owing to differences in operationalization and scale (Wutich et al. 2020b). Water needs are approached as a physiological and nutritional question (Sawka et al. 2005) and examined on a more acute scale (hourly or daily). In contrast, WI is studied as an experiential or resource-based question (Jepson et al. 2017, Octavianti & Staddon 2021) and examined on a longer timescale (month or year). Yet how people meet both their immediate physiological water needs and the water needs for all of life's tasks, including cooking, hygiene, and livelihoods, is inherently connected and tied to human biological outcomes. Competing effects of hydration and stress along with ultimate and proximate causation related to coping strategies lead to complexities in deciphering how water needs versus WI affect human biology.

In this review, I aim to connect these literatures and discuss their intersecting pathways to affect human biology and health. First, I describe advances in our understanding of human water needs, thirst, and hydration strategies from an evolutionary and biocultural perspective. Second, I critically appraise recent advances in the WI literature and in how humans cope in response to WI. Third, I use the embodiment framework to discuss how human water needs entangle with WI to affect human biology through coping strategies and changes to hydration strategies. I close with a discussion of how lifestyle changes and climate change serve to improve or disrupt and threaten water access and have their own compounding effects on human biology.

WATER NEEDS, THIRST, AND HYDRATION STRATEGIES

From prior to conception to older life, water and proper hydration support a healthy life (Rosinger 2020). While recommendations exist for daily adequate water intake (e.g., 3.7 L for men; 2.7 L for nonpregnant, nonlactating women, 3.0 L for pregnant women, and 3.8 L for women who are lactating), these are not hard-and-fast rules (Food Nutr. Board Inst. Med. 2004). Water needs depend on sex, age, reproductive status, body size and composition, physical activity, lifestyle, health status and medication use, and environmental conditions (e.g., altitude, heat, humidity, solar radiation, and climate control) (Rosinger 2021, Sawka et al. 2005, Yamada et al. 2022). Other factors, such as menstruation cycle and changes in hormones, may also affect water intake but have less evidence to date (Giersch et al. 2020). Ensuring that water needs are met is important particularly for infants and children, older adults, pregnant and breastfeeding women, and workers who are highly susceptible to dehydration (e.g., sugar cane workers, firefighters) in extreme environments (see Ocobock 2023, this volume, for more on adaptations to extreme environments). Yet to understand human biological variation in water needs, it is critical to examine our evolutionary past, the adaptations that influence water needs and thirst, and early-life environments and exposures.

Water insecurity

(WI): the inability to reliably access, benefit from, and use water for a healthy lifestyle

Coping strategy:

the behavioral or technological adaptation or response to obtain water when facing water insecurity

Human water needs:

the physiological requirements needed to maintain homeostasis and cellular function

Thirst:

the mechanism that drives the behavioral response to drink and maintain body fluid balance

Hydration strategies:

the biocultural responses to meet one's water needs from available environmental water and food resources

Embodiment:

the process through which social and ecological environments become incorporated into physioanatomic characteristics that influence health

Human adaptations to heat are a key evolutionary driver that shaped variation in human water needs (Lieberman 2015). Body form and composition changes between *Australopithecus* and *Homo erectus* resulted in changes in exposure to solar radiation and wind, thus reducing water needs by 15–18% (Wheeler 1993). Human hair and scalp hair evolution may also be a thermoregulatory response with implications for body water conservation. Tightly curled hair in high solar radiation and heat conditions helps maximize cooling, thus reducing water losses by decreasing the need for evaporative cooling (sweat), particularly in dry conditions (Lasisi 2021). Other adaptations, such as eccrine sweat glands and the *Engrailed 1* gene mutations, allow for enhanced cooling ability in response to heat and provide the potential to lose copious amounts of water (Aldea et al. 2021, Best et al. 2019). The kidneys, which are unique among mammals, serve as a critical bridge between water conservation adaptations and thermoregulatory adaptations to maintain homeostasis (Beuchat 1990). Selection pressure on a kidney expressed gene, *STC1*, among Turkana pastoralists in response to dehydration and meat consumption has been associated with urea levels (Lea et al. 2023). Thus, extreme heat, aridity, and WI may create selection pressures with implications for genetic adaptations for water needs.

Overall, human water turnover is 30–50% lower than that of apes adjusted for body size (Pontzer et al. 2021). This finding may be due to variation in hydration strategies where apes consume a greater percentage of water from foods with preformed water and have a higher water-to-energy ratio. This differential hydration strategy may lead to a secondary effect on anticipatory thirst, which occurs in response to impending changes in hydration, such as dietary intake, sleep, and heat stress. More work with nonhuman primates is needed to untangle this difference in water needs and hydration strategies between humans and nonhuman primates (Nelson et al. 2022, Wessling et al. 2018).

Interactions with lifestyle, environment, and subsistence patterns mean that people can have two times the water needs as those of others (Swanson & Pontzer 2020). A developmental origins of health and disease (DOHaD) framework provides a model for understanding how exposures to stress in utero and in early life can affect later-life health (Mulligan 2016). The DOHaD of water needs hypothesizes that environmental exposures can help calibrate individual water needs to local ecologies (Rosinger 2020). Exposures to water scarcity and extreme heat in utero can reduce birth weight and increase risk of preterm birth, which have critical implications for water needs and kidney size (Hoy et al. 1999, Randell et al. 2020). These exposures to heat, water availability, and dehydration may shape water needs through effects on body size, kidney mass, number of nephrons, and shifts in thirst set points (Hoy et al. 1999, Rosinger 2021, Ross & Desai 2005). Yet, a key neglected factor in studies examining water needs has been how the environment and WI may calibrate individual thirst perception.

Thirst

Intra- and interpopulation patterns of water intake show large variation with individuals who consume far less and far more than the median amount (Rosinger 2020). Thus, other factors, such as priming of water needs and thirst, are likely at play in driving water intake. Two types of thirst exist: homeostatic thirst, which occurs in response to a water deficit, and anticipatory thirst, defined above (Gizowski & Bourque 2018). The two primary signals for thirst are increased plasma osmolality (sensed by osmoreceptors in the hypothalamus) and decreased blood volume (sensed by low-pressure baroreceptors in the venous system leading back to the heart) (Gizowski & Bourque 2018). In day-to-day life, the former predominates and thirst follows vasopressin release with a threshold-gain type response pattern.

Two adaptations may alter thirst. One adaptation would be a threshold shift in the vasopressin–plasma osmolality relation, i.e., turning on thirst earlier or later. The second would be an elevated

Developmental origins of health and disease (DOHaD):

how early-life exposures affect disease risk later in life, oftentimes through epigenetic changes

Vasopressin: the antidiuretic hormone responsible for body water homeostasis

Hydration status: the internal state of water balance within an organism

response (higher slope) once thirst is initiated. Early-life environmental exposures to water scarcity may create one of these adaptations (Rosinger 2020).

While a body of work has described various factors affecting thirst and how conditions such as diabetes insipidus and medication use alter thirst (Carroll 2020, Rolls et al. 1982), little is known about thirst perceptions outside of Western populations and how experiences surrounding WI may affect these set points. The only study to date investigated how thirst perception varied across hot-dry and hot-humid environments and how early-life environments affected thirst perception (Rosinger et al. 2022a). In these extreme thermal environments without access to climate control, thirst was not strongly tied to ad libitum hydration status. Furthermore, we found that a significant interaction existed between season of birth and ambient temperature on a proxy of thirst perception—pleasantness of drinking water. Daasanach pastoralists from northern Kenya born in the wet season (meaning they were in utero during less water availability) had blunted pleasantness of drinking water at higher temperatures, a situation of greater water needs. With hotter temperatures, their desire to drink water was lower than those who were in utero during greater water availability. The adaptive thirst response may be blunted among those who experience less water in utero (Ross & Desai 2005). More work is needed to understand the role of water stress and WI during gestation in priming thirst, as well as about cross-cultural variation in interoceptive accuracy and sensitivity to thirst and dehydration.

Human Adaptations to Water Needs: Hydration Strategies

Beyond the amount of water that people consume and the thirst that drives intake, hydration strategies reveal *how* people meet their water needs (Rosinger & Tanner 2015). Hydration strategies are a nutritional adaptation used to fend off the stress, both physiological (dehydration) and psychological (worrying about water), of water needs (Stinson 1992). Across populations, hydration strategies are flexible, similar to diet. They depend on access to safe water, beverages, and food resources, as well as income, and are influenced by embedded cultural traditions and preferences in a given physical, social, and structural environment.

The Institute of Medicine (Food Nutr. Board Inst. Med. 2004) and the European Food Safety Authority (EFSA 2010) state that food contributes ~20% and 20–30% of water intake, respectively. Yet when hydration strategies are examined cross-culturally, far more variation emerges in both total intake and the percent of water coming from food sources (**Figure 1**).

The percent of water intake from foods ranges from 18.3% in the United States (Kant et al. 2009) to ~50% among 'Tsimane' forager-horticulturalists in Bolivia (Rosinger & Tanner 2015) and in Japanese (Tani et al. 2015) and Chinese (Zhang et al. 2019) samples. Dietary traditions of soup and rice and water-rich foods, for example, in Asian and South American cultures lead to higher water intake from food sources. By contrast, higher intake of processed, less water-dense foods along with greater diversity of marketing for beverages may lead to higher fluid intake in Western, high-income countries. The nutrition transition has affected dietary patterns globally, which has also had implications for hydration strategies through effects on preferences, usage of local water sources for other beverages, and water availability and WI. See the sidebar titled Dietary Methods.

WATER INSECURITY AND COPING STRATEGIES

WI differs from physiological water needs because it covers the water needed for all of life's tasks and livelihoods. In-depth descriptions of WI literatures are available elsewhere (Jepson et al. 2017, Young et al. 2021b). Briefly, WI has four main dimensions: availability (physical presence of water), accessibility (ability to get to and use the available water without physical, economic,

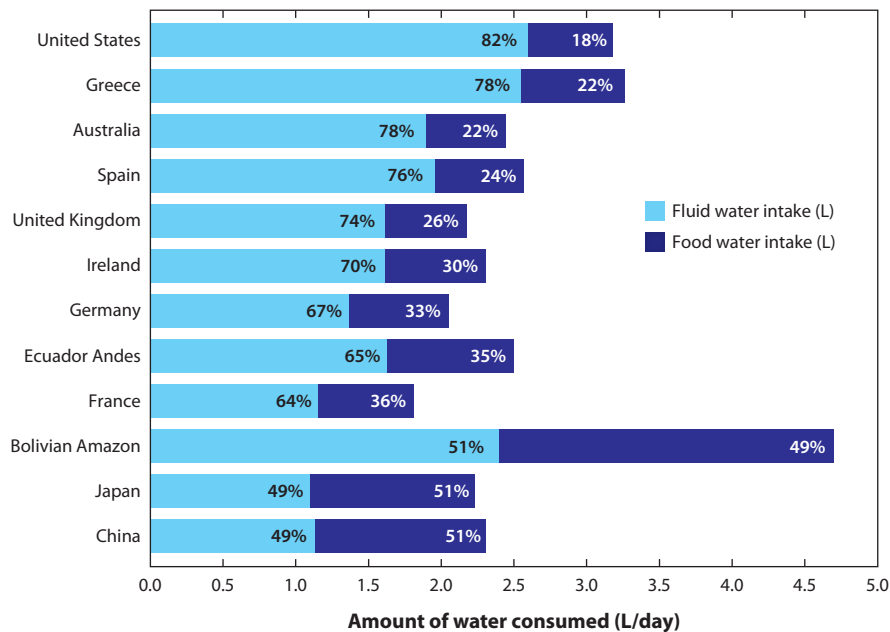


Figure 1

Cross-cultural variation in hydration strategies illustrates differences in the amount and proportion of daily water consumed from foods (*dark blue*) compared with water consumed from fluids (*light blue*) among adults. Men's and women's intakes are combined when sex was reported separately; the data provided are the author's own calculations. Data from Athanasatou et al. (2016), Berti & Leonard (1998), Guelinckx et al. (2016), Kant et al. (2009), Malisova et al. (2016), Manz et al. (2012), O'Connor et al. (2014), Rosinger & Tanner (2015), Tani et al. (2015), Zhang et al. (2019).

cultural, or political barriers), use (whether the available and accessible water is also acceptable and safe for drinking and for nonconsumptive purposes), and stability (how reliable the availability, accessibility, and use are over time) (Young et al. 2021b). WI results when any domain is compromised.

Water Insecurity Experiences

The framework of WI provides a way to study lived experiences surrounding water, including inequities in water access and quality, affordability, environmental injustice, coping strategies, and

DIETARY METHODS

Hydration strategies are estimated through 24-h multiple-pass dietary recalls. While self-report is subject to recall bias, population means from dietary recalls are consistent and slightly lower than when water turnover is calculated with methods of deuterium in doubly labeled water, which may be due in part to metabolic water created by food consumption (Swanson & Pontzer 2020). For example, in the United States, the mean daily water intake from dietary recalls compared with deuterium analyses was calculated at 3.46 versus 3.59 L for men and 2.75 versus 2.99 L for women, respectively. Yet data about the sources of dietary water are essential to inquire about because they provide implications for infection risk, microbial diversity, and energy consumption.

their outcomes. This framework allows for a deeper examination of several key questions: (a) how big a problem WI is within and across sites (Young et al. 2022); (b) how experiences of different dimensions of WI are affected by climatic and hydrologic changes as well as shocks from extreme climatic events and anthropogenic factors (Broyles et al. 2022, Rosinger et al. 2023); (c) how WI intersects with other insecurities, such as food, housing, and energy, to compound vulnerabilities and how it affects livelihoods (Shah 2021, Wutich & Brewis 2014); (d) how marginalized groups differentially experience WI and how this variation affects water-related perceptions, beliefs, and coping behaviors (Jepson 2014); and ultimately (e) how WI is associated with well-being, health, nutrition, and human biology (Rosinger & Young 2020b).

A critical development in the WI literature has been the cross-cultural validation of household-level WI experiences (HWISE) scale over the prior four weeks (Young et al. 2019b) and the individual-level WI experiences (IWISE) scale over the prior year (Young et al. 2021a). To capture all dimensions of WI, complementary measures to scales provide added nuance (Jepson et al. 2021). Perceptual factors, such as trust and trustworthiness of available water sources (Wilson et al. 2022), tap water avoidance (Rosinger 2022), and cultural relationships with water (Duignan et al. 2022), serve as important indicators of water use and historical relationships with water sources and the provision of clean water. Furthermore, additional data on objective water quality, assessments of access to clean water, affordability and cost (both money and time) of getting water, prior experiences with major water crises, and perception of trust and water quality will aid in this complementary effort to create a holistic understanding of WI (Boateng et al. 2018, Rosinger et al. 2021c).

Ecologies interact with how people experience and perceive water-related problems. Hydrological and climatic variation affect dimensions of WI differentially across regions. Some dimensions such as availability may improve during the rainy season, but others (access or use) may worsen (Broyles et al. 2022). While water scarcity has deservedly received most of the attention in WI research, an increasing body of literature is demonstrating that WI occurs in unexpected environments, such as in water-rich regions. For example, in the Amazon, dimensions of WI pertain more to flooding or too much water as well as water cleanliness and pollution rather than insufficiency (Rosinger 2018, Tallman 2019).

Similarly, WI can occur in high-income countries such as the United States, where access to clean water is theoretically universal. Yet one percent (two million people) do not have access to piped water, while a further 20% do not drink their tap water, with tap water avoidance (and stark racial/ethnic disparities) increasing in the United States since the Flint, Michigan, water crisis, signifying distrust and insecurity (Rosinger et al. 2022b). Issues of access, distribution, contamination, aging infrastructure, and environmental injustice color water problems in North America (Switzer & Teodoro 2018). These multiple intersecting marginalities affect places such as the colonias on the Mexico–Texas border (Jepson 2014, Wutich et al. 2022a), the US South (Nwanaji-Enwerem & Casey 2022), older historically Black cities such as Detroit and Baltimore (Gaber et al. 2021, Swain et al. 2020), Native American reservations (Doyle et al. 2018, Schell 2020), rural towns relying on well water (Mulhern et al. 2022), and low-income mobile-home communities (Pierce & Jimenez 2015). These disparities in clean water, infrastructure, and distrust influence what people deem safe to drink and the measures they must take to meet their water needs.

Human Adaptations to Water Insecurity: Coping Strategies

The anthropological literature examining responses and resiliency to WI is increasingly important (Achore et al. 2020, Venkataramanan et al. 2020a). When a water source is distant or flows intermittently, this variability forces behavioral changes, such as fetching water at specific times or leaving taps on to capture and store water for future use.

Coping strategies include any way to mitigate WI, including treating and filtering water; transferring water between households, such as lending/giving and borrowing/receiving (Brewis et al. 2019b); using multiple sources (Elliott et al. 2017); storing large quantities; buying alternative sources such as bottled water (Radonic & Jacob 2021); hauling water; stealing water; rainwater harvesting; digging/building new sources (Achore et al. 2020); substituting alternative sources, such as traditionally fermented beverages or other beverages; restricting use for hygiene/cleaning; reducing intake, e.g., among migrants traversing the Sonoran Desert (De León 2013); shifting individual use to buffer others, such as children (Maxfield 2020); and relocating/migrating (Stoler et al. 2021).

A cultural hydration strategy may influence one's decision to engage in a coping strategy for WI. For example, in a population with tea- or coffee-drinking customs, such as Daasanach pastoralists, individuals will borrow water from a neighbor if they do not have enough in the house to make tea (Ford et al. 2023). Conversely, the ability to engage in coping strategies influences which hydration strategies people use. Those without access to clean water who engage in coping strategies and modify their hydration strategies may suffer disproportionately (Rosinger et al. 2023). While these coping strategies may provide the water needed in the short term, they may have unintended consequences or create extra mental burdens, which increases the stress associated with WI.

THE INTERSECTION OF WATER NEEDS AND WATER INSECURITY WITH HUMAN BIOLOGY

The environment of “too little,” “too much,” and “too dirty” water can get imprinted on the body and mind (Krieger 2005, Worthman & Kohrt 2005). A significant amount of work has examined biological effects of dehydration (Rosinger 2020), and separately the psychosocial distress of WI, including “suffering for” and “suffering from” water (Ennis-McMillan 2001, Sultana 2011, Workman et al. 2021, Wutich et al. 2020a). Embodiment provides a framework for understanding how those experiences—and the lengths to which people go to meet water needs—“get under the skin” to affect human biology. Yet a key challenge remains in teasing apart the embodiment of water needs from WI as well as from the proximate hydration or coping strategies used to meet water needs and mitigate WI at various life stages (**Figure 2**).

Embodiment of Water Needs and Water Insecurity

The embodiment of water needs and WI begins at conception and is intimately tied to the nutritional environment as water is critical to growing food, cooking, and livelihoods. An increasing body of evidence supports the connection between WI and food insecurity and is covered elsewhere in depth (Miller et al. 2021, Rosinger & Young 2020b, Wutich & Brewis 2014, Young et al. 2021b); therefore, here, I highlight the connection to maternal and early-childhood nutrition since those early-life experiences—both of hydration and of WI—are critical to growth, illnesses, thirst, and lifelong embodiment of water needs. For women who are pregnant, water restriction and inadequate hydration may have implications for birth outcomes, including preterm birth and lower birth weight (Rosinger 2020, Rosinger et al. 2021b). WI complicates infant and childhood feeding practices (Schuster et al. 2020). When dealing with WI, mothers report breast milk insufficiency and children's risk of dehydration and diarrhea (Schuster et al. 2020). This concern is particularly salient in areas with water contamination or scarcity because the replacement for breast milk is formula, which must be mixed with unclean water and may lead to diarrheal diseases and malnutrition (Collins et al. 2019). While poor water access adversely affects early-life dietary diversity (Choudhary et al. 2021), little evidence currently exists that WI independently affects growth during infancy, though it is possible that coping strategies that mitigate the stress from WI mask the effect (Koyraty et al. 2022).

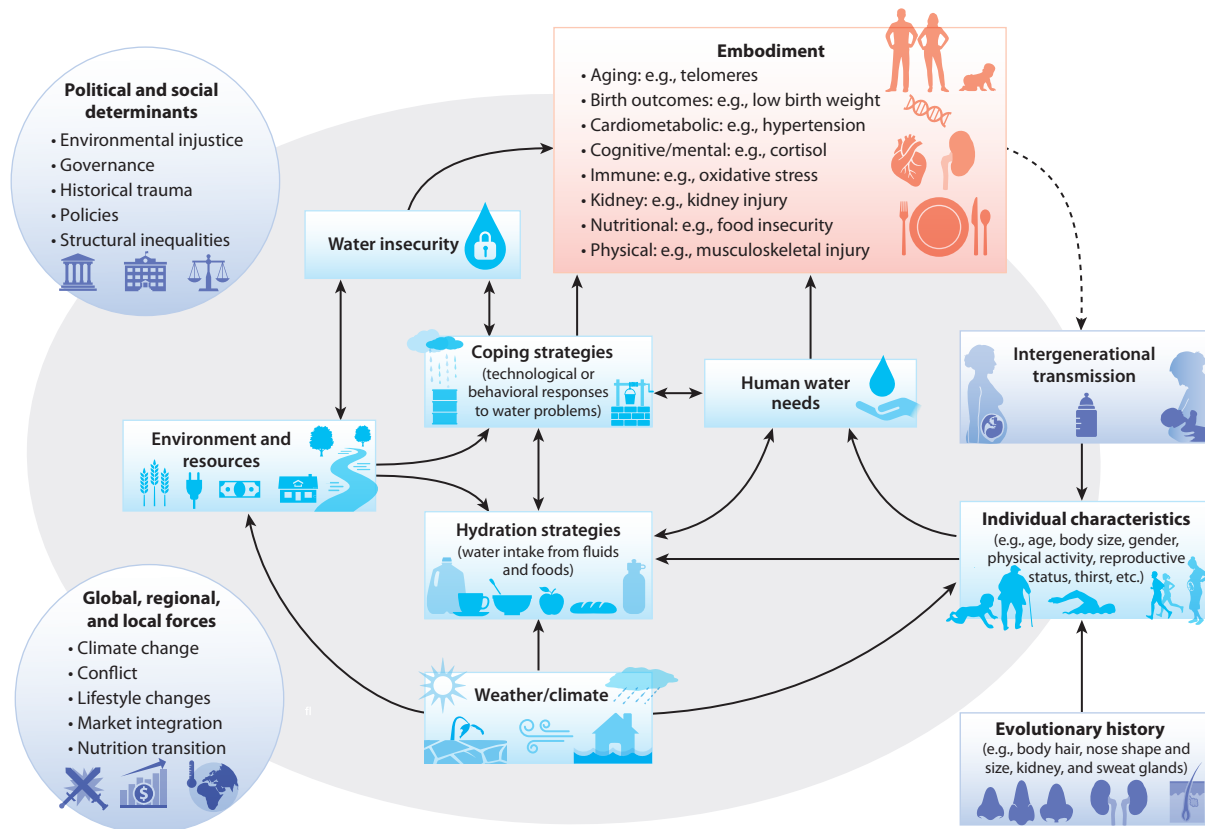


Figure 2

Conceptual model integrating the ultimate and proximate drivers and human biological consequences of human water needs, water insecurity, and the coping and hydration strategies to meet those needs. Outer dark blue circles represent the ultimate forces shaping structural environments and exposures; the dark blue boxes represent the evolutionary and intergenerational forces shaping individual variation and population adaptations; the light blue boxes represent the proximate drivers shaping exposures and variation; the dark red box represents the various ways that water needs, water insecurity, and coping and hydration strategies are embodied in health consequences; the dashed line indicates potential intergenerational transmission of the embodiment on future individual variation; and the outer gray circle illustrates that human water needs are enmeshed in a cycle of evolutionary, intergenerational, structural, climatic, and anthropogenic factors.

Telomere: a region of repetitive DNA sequences at the end of a chromosome; telomere length is used as a biological marker of age

However, combined nutrition and water, sanitation, and hygiene (N+WSH) interventions demonstrate a clear connection to infant growth and development of the hypothalamic-pituitary-adrenal (HPA) axis supporting the stress response. In one study, this intervention was associated with accelerated telomere reduction in the first year of life after intervention followed by telomere increases in the second year, due to better growth and fewer diarrheal and parasitic infections during the first year rather than accumulated stress (Lin et al. 2017). In a second study, this N+WSH intervention was associated with improvements in physiological measures of the set point, regulation, and reactivity of the HPA axis for children in their first two years of life (Lin et al. 2021).

The physiological effects of hydration make it hard to untangle associations between WI and stress if hydration status is not concurrently measured. Having chronically low water intake is associated with greater urine osmolality, urine specific gravity, vasopressin, and increases in the stress hormone cortisol (Perrier et al. 2013a, Rosinger 2020). For example, dehydration increases

blood cortisol (Perrier et al. 2013b), yet hair cortisol (a chronic stress biomarker) has also been associated with inequities in access to clean water, though not WI scores (Rosinger et al. 2021c). Blood pressure (BP) is frequently used as an indicator of stress but is influenced by hydration and environmental factors. This physiological process helps explain why there has been mixed evidence to date on the association between WI and BP (Brewis et al. 2019a, Tallman et al. 2022). Dehydration and hotter temperatures both lower BP, yet WI may increase risk of dehydration (Rosinger 2018). Thus, examining and controlling for hydration status in the relation between WI and stress biomarkers—along with environmental variables such as heat stress, which also affect BP, water needs, and hydration status—is critical.

Advances have begun to untangle the effects of water stress on aging and the immune system. Among older US adults, higher water intake and improved hydration levels were associated with lower circulating triglycerides and low-density lipoprotein cholesterol and higher high-density lipoprotein cholesterol, but they were unrelated to markers of inflammation [C-reactive protein (CRP)] (Jacques et al. 2021). Similarly, WI was not associated with markers of inflammation (CRP, Epstein-Barr virus) in the Amazon (Tallman et al. 2022). In terms of immune function, evidence from animals indicates that chronic water restriction and heat stress lead to telomere shortening and oxidative stress (Dupoué et al. 2020). Among humans, ad libitum dehydration was also associated with significantly lower stimulated cytokine production among healthy middle- to older-aged adults, indicating potential suppressed immune function (Davis et al. 2023). During exercise, dehydration and hyperthermia increase cellular stress and oxidative stress (Hillman et al. 2011). Yet while oxidative stress and cellular damage increase during dehydration, they are then repaired with fluid consumption following exercise to exhaustion, indicating that the embodiment of acute dehydration may be transient (Paik et al. 2009). However, longitudinal data indicate that chronic dehydration is associated with shorter life span, degenerative diseases, and increased kidney injuries (Allen et al. 2019).

Dehydration and WI both have implications for cognitive function. Dehydration leads to the deterioration of sustained attention tasks among kids—particularly without climate control (Bar-David et al. 2009)—and adults (Bethancourt et al. 2020). An interesting connection exists between cognitive function and water-related anxiety that vulnerable groups face, which may be exacerbated by heat. Mild dehydration increases thirst and can increase anxiety surrounding water access (Young et al. 2019a). Older adults are particularly vulnerable to both dehydration and WI as they suffer a decoupling of thirst from hydration status, and effects on cognition for them may have increased implications for coping strategies and risk of other embodied effects.

Embodiment of Coping and Hydration Strategies

An added complication for understanding the embodiment of water is that most WI scales include questions that capture coping strategies to conditions of water challenges (e.g., reducing water use or changing plans) (Hadley & Freeman 2016, Young et al. 2019b) or even health outcomes (e.g., getting hurt fetching water) (Rosinger 2018). This mix of experiences of water problems (e.g., worrying about water) with responses and health outcomes increases the complexity of whether associated health outcomes are due to WI or to the proximate coping strategies. For example, WI experiences through exposure to contaminated water can increase risk of diarrheal infection, dehydration, and psychosocial stress (Rosinger & Young 2020b), but borrowing water to meet needs may yield water that is unclean or that creates stressful obligations (Stoler et al. 2019, Wutich et al. 2022b).

Another example of this complexity is in examining the embodiment of WI in chronic pain, injuries, and gender-based violence, which is a product both of the environment and of specific coping strategies (Adams et al. 2022, Tallman et al. 2023). In areas without household piped

Oxidative stress:
an imbalance in
oxygen reactive species
in cells and tissues and
the ability to detoxify
these reactive products

water, a key coping strategy to WI is water fetching. Yet water fetching places individuals, most often women, at greater risk of getting injured and is particularly dangerous for pregnant and postpartum women owing to the risk of spontaneous abortions and uterine prolapse, respectively (Meierhofer et al. 2022, Venkataramanan et al. 2020b). Reports of injuries and pain are increasingly common for those fetching water, and thus their bodies mirror the stories they tell regarding WI, experiences that can be understood as “suffering from water” (Sultana 2011). Yet the stress they accumulate from the injuries may not necessarily be visible. We tested how WI was embodied in lowland Bolivia. We found that women, but not men, who reported a household water-related injury had 36% higher hair cortisol concentration and that unimproved water sources were associated with higher stress (Rosinger et al. 2021c). This finding demonstrated a gendered embodiment of WI among those responsible for supplying household water and engaging in the coping strategy of water fetching. Thus, it is critical to measure coping strategies and experiences surrounding water acquisition along with self-reported health measures and objective biomarkers to unveil the full toll of WI.

Another ubiquitous coping strategy to deal with WI is water sharing, or household-to-household water transfers (Wutich et al. 2018). This strategy of acquiring water, often from neighbors, is universal in sites in low- and middle-income countries (LMICs) that deal with water problems (Rosinger et al. 2020). Yet this coping strategy is not without cost. Borrowing and lending water is a distressing form of reciprocity associated with greater reports of being ashamed, upset, angry, and in conflict with neighbors (Wutich et al. 2022b). However, the perception of unfairness matters when examining mental health outcomes in relation to WI and water borrowing (Brewis et al. 2021). Thus, to acquire water, often as little as 3–5 L, just enough to cook a meal, make tea, or have some water to drink, may create an obligation and future problems (Wutich & Ragsdale 2008). Yet whether borrowing water negatively affects stress may also depend on the level of one’s participation in water-sharing networks. In northern Kenya among Daasanach seminomadic pastoralists, the level of water borrowing moderated the relationship between WI and psychosocial stress (Ford et al. 2023). While greater WI was associated with higher perceived stress, this finding was only the case for those who rarely or never borrowed water; whereas for those who borrowed water frequently, as WI increased, perceived stress did not increase. Thus, it is important to understand context, ecology, perception of fairness, and cultural history of coping strategies to WI and their ties to mental and physical health outcomes.

In different waterscapes, variation in WI and trust of water sources—and resulting embodiment—are shaped by interactions between coping and hydration strategies. Tap water avoidance signals underlying distrust of the available water source and is aimed at protecting oneself or buffering one’s children from potentially contaminated water. This practice leads to changes in hydration strategies, or substitutions, aimed at mitigating perceived uncleanness of water sources, which have the potential to be embodied in health (Rosinger 2022). For example, in Bolivia, Tsimane’ who consumed more of their water from food sources relative to plain water, which was contaminated with pathogenic bacteria, had significantly lower probability of gastrointestinal illness (Rosinger & Tanner 2015). In the United States, distrust leads to the coping strategy of buying alternative sources, which results in shifts to hydration strategies such as bottled water and sugar-sweetened beverages (Bradley et al. 2023, Javidi & Pierce 2018). This coping strategy has both socioeconomic implications (bottled water and sugary drinks are both more expensive than tap water) and human biology implications (exposure to contaminants, sugar, and stress surrounding finances). Higher sugary drink intake leads to an increased risk of rapid weight gain (Schwartz et al. 2016), diabetes (Twarog et al. 2020), cavities, and cardiometabolic disease (Malik 2017) among children and adults. Furthermore, hydration strategies with higher sugary drink intake are associated with shorter telomeres (Leung et al. 2014).

Variation in hydration strategies and coping strategies related to changes in water source use has further implications for the microbiome and nutrition. For example, *chicha* (a traditional fermented beer often consumed in a communal gourd) intake was associated with greater oral microbiome diversity among the Shuar in Ecuador (Colehour et al. 2014). Thus, the hydration strategy of drinking *chicha* has both hydration and microbiome consequences (Rosinger & Bethancourt 2020). However, even variation in plain water sources through coping strategies of water storage or reliance on alternative sources can affect the gut microbiome. For example, in Ecuador, children with worse stored household water quality had lower gut microbial diversity (Piperata et al. 2020), whereas in the United States, adults who relied on well water had significantly greater fecal microbial diversity than did those who relied on city, bottled, or filtered tap water (Vanhaecke et al. 2022). For more on the microbiome, see the review by Amato & Carmody (2023, this volume).

The consequences of coping and hydration strategies disproportionately affect marginalized groups. For example, in the United States, Black and Hispanic children and adults are more likely to avoid tap water (Rosinger et al. 2022b, Rosinger & Young 2020a) and to drink sugary drinks (Elfassy et al. 2019) than are their non-Hispanic white and Asian counterparts, leading to higher risks of negative cardiometabolic health problems. Low-income children also have higher sugary drink and bottled water intake and lower tap water intake relative to higher-income children (Elfassy et al. 2019, Rosinger & Young 2020a). Overall, the embodiment of changes in coping and hydration strategies in response to WI is distributed unequally across contexts as suffering and water access problems are greater in marginalized groups and LMICs.

Disrupting Forces: Lifestyle Changes and Climate Change

Social changes through market integration, nutritional transitions, and urbanization along with extreme climatic events have distinct implications for human water needs, WI, and hydration and coping strategies, particularly in rural and LMIC populations. These lifestyle and ecological changes affect the availability of water, beverage, and food sources and are coupled with increased demand and pollution, thus changing the cost of water and what is appropriate to consume. Here I highlight a few examples to demonstrate how these changes have implications for the embodiment of water.

Lifestyle changes can disrupt hydration strategies that serve to clean water such as local beverages used as adaptive responses to the ecology. One example is the hydration strategy of fermented beverages in areas where water safety is an issue. High *chicha* consumption in response to a lack of access to clean water was pointed to as a key reason that Ecuadorian Shuar men and women had very high water turnover (9.4 and 4.8 L per day, respectively) (Christopher et al. 2019, Harner 1984). Similarly, Tsimane' have a long tradition of *chicha* consumption. However, missionaries condemned *chicha* because it contained alcohol, resulting in changes to the way it is made and less reliance on it as a beverage. Now, *chicha* is consumed most frequently before it is fermented, thus negating the lysing properties of fermentation (Rosinger & Bethancourt 2020).

Other lifestyle changes (e.g., sedentarization, market integration) have implications for hydration strategies and health. For example, compared with nomadic Turkana pastoralists in Kenya, settled Turkana had reduced access to milk supply, lower milk intake, and worse reported health (Barkey et al. 2001). Similarly, Daasanach seminomadic pastoralists in Kenya who were more sedentary and did not consume milk daily had an increased probability of hypertension and poor kidney function due to their higher reliance on saltier water sources (Rosinger et al. 2021a). Access to water sources and beverages is affected not only by market integration, but also by perceptions. When outside entities change infrastructure and communicate messages that local water sources

are not clean but these entities also do not provide improved water sources, this situation creates a dissonance and increased dissatisfaction among local populations with water, which may increase stress (Rosinger et al. 2021c).

Climate change and extreme climatic events intersect with social and livelihood changes to disrupt human water needs, WI, and the coping and hydration strategies used to meet those needs. Extreme climatic events, such as extreme storms, flooding, drought, and heat waves, create shocks that disrupt water and food access, necessitating coping strategies and alteration of hydration strategies (Rosinger 2023). For example, during Hurricane Maria, residents in Puerto Rico had to form water-sharing networks to provide water to neighbors after power and water outages (Roque et al. 2021), and this practice may have affected hydration strategies and health. In lowland Bolivia, a historic flood in 2014 killed crops in horticultural fields; as a result, *chicha* consumption declined and Tsimane' began mixing raw water with sugar or fruit instead (Rosinger et al. 2023). In the Horn of Africa, extreme drought and flooding killed hundreds of thousands of livestock, resulting in lower access to milk for calories and hydration (Matanó et al. 2022). Hotter temperatures and heat waves decrease water availability via evaporation, but they also affect electrical grids that allow water pumps to run (Jasechko & Perrone 2020). This change in access, with wells running dry, necessitates the use of coping strategies, such as shifting to buying bottled water or sugary drinks. It is further of note that human limits to heat are lower than previously thought as the 35°C wet-bulb globe temperature limit has been revised down to 32°C, indicating that humans are less able to compensate in humid heat without resulting health complications (Wolf et al. 2023). While extreme heat increases dehydration and hyperthermia (and the associated embodiment of dehydration discussed above), rehydrating with sugary beverages following dehydration and heavy exercise increases kidney injury and can lead to chronic kidney disease (CKD) (Chapman et al. 2019, Johnson et al. 2019, Sato et al. 2019). Thus, future climate scenarios may lead to rising CKD rates if hydration strategies are not carefully considered.

Climate and anthropogenic effects on ingested water quality can be a source of WI and can be embodied. Changes in water temperature and water quality lead to a greater risk of enteric infections, affecting nutrient delivery and gut microbiome diversity (Guerrant et al. 2013, Karn & Sharma 2021). These changes in the microbiome may dysregulate cardiometabolic pathways and lead to a higher risk of stunting and obesity (Thompson et al. 2020). Furthermore, climate change-related fluctuations in rainfall patterns and rising temperatures affect water quality concentrations through subsequent droughts, floods, dilution, and runoff. Extreme climatic events such as flooding may then lead to contamination of water supplies when encountering waste or chemicals. With greater water scarcity and flooding events, humans will encounter greater exposures to contaminants such as arsenic, *Escherichia coli*, fluoride, lead, nitrates, and per- and polyfluoroalkyl substances (PFAS), which have important health and human biology implications and are covered elsewhere in depth (Fewtrell & Bartram 2001). In particular, PFAS are an emergent and relatively novel ecological stressor and contaminant of human concern, as they are an endocrine-disrupting chemical; there is growing evidence of their detrimental effects on a suite of human biological outcomes, including cardiometabolic health, kidney health, and cancer (Fenton et al. 2021). In the United States, PFAS have been found to be inversely associated with biological aging and oxidative stress, indicating deceleration of the aging process, at least at low circulating levels (Chaney & Wiley 2023). It is noteworthy that the US Environmental Protection Agency in 2023 proposed maximum contaminant levels for 6 PFAS, most notably perfluorooctanoic acid and perfluorooctanesulfonic acid at 4 parts per trillion.

Another water quality concern is salinity, which is a key integration of water needs and WI affected by climate change and anthropogenic effects and illustrates the implications of water quality

composition (Bradley et al. 2023). Increased salinization of water through seawater intrusion, road salting, water softeners, evaporation, and other sources has critical implications for human biology through water needs, WI, and the water sources that people use. Water salinity negatively affects kidney health and cardiovascular health, with evidence from Bangladesh (Khan et al. 2014, Scheelbeek et al. 2017) and Kenya (Rosinger et al. 2021a). Yet the composition of salt is embodied differently depending on the ionic concentration. When salt is predominantly composed of sodium and chloride, the potential harm to kidney and cardiovascular health is greater than when it includes calcium, magnesium, and potassium, which are protective against hypertension (Naser et al. 2020). Yet the salt content in water also affects water retention and body water homeostasis (Rakova et al. 2017). Freshwater salinization not only increases exposures from the accumulation of sodium and chloride in water supplies, but also has a suite of negative ecological implications, including the release or mobilization of trace metals, acidification, and increases in organic matter (Kaushal et al. 2022). Thus, the resulting chemical cocktail in freshwater may have unintended health consequences in the future.

CONCLUSIONS

Understanding how water needs and WI intersect to affect human biology requires an integrative approach that also considers how people meet their water needs through hydration and coping strategies. The divergence from human water needs, alteration of hydration strategies, and a reliance on coping strategies can each have interlinked effects (both good or bad depending on context) on psychosocial and physiological stress and can affect exposures and health outcomes. These experiences can affect future generations through experiences in utero, which may shape thirst set points and nutritional trajectories. Those factors prime an individual for a set of environmental resources. Within this cycle, social, structural, and climatic changes have additional influences on water availability and quality, on perceptions of what is appropriate to consume, and on required hydration and coping strategies, which further affect human biology. It is at this intersection—considering our evolutionary past, scale, and linkage of experiences, exposures, responses, and outcomes—where a complete picture emerges of the human biology of water.

FUTURE ISSUES

1. Greater attention to human adaptability and both behavioral and genetic adaptations to diverse environments is needed for studying water needs and water insecurity.
2. An important way forward to study human water needs will be pairing deuterium with dietary recall to assess hydration strategies and water turnover in concert with hydration biomarkers.
3. Developmental and intergenerational perspectives, as well as novel biomarkers to better test how environmental stressors affect hydration status and water insecurity and relate to embodiment, will be critical moving forward.
4. How do different dimensions of water insecurity experiences affect perception and corresponding coping strategies resulting in variation of exposures?
5. How do water- and climate-related experiences and shocks during sensitive periods affect methylation changes in the epigenome and cortisol responses that may lead to cross-cultural variation in thirst and the embodiment of water needs and insecurity?

6. Can integrative interventions that improve structures, infrastructure, water, and nutrition mitigate the embodiment of water insecurity?
7. High-resolution data on exposures and early-life experiences of climate change, heat stress, lifestyle changes, policies, and structural inequalities are needed to better link to human biology outcomes.
8. Future work should examine the implications of changes in water quality driven by climate change and anthropogenic factors for water needs and human biology.

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